Practice Problem

Saturday, 23 May 2026 12:45 am

Solve(Any two)

Determine the rank of the given matrix.

(a) $\begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$
Answer: 2

(b) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 4 \\ 1 & 4 & 1 \end{bmatrix}$
Answer: 3

(c) $\begin{bmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 1 & 0 & 5 & 1 \\ 2 & 1 & \frac{2}{3} & 3 & \frac{1}{3} \\ 6 & 6 & 6 & 12 & 0 \end{bmatrix}$
Answer: 3

Determine whether the given set of vectors in $\mathbb{R}^n$ is linearly dependent or linearly independent.

(a) $\mathbf{v}_1 = (1,2,3)$, $\mathbf{v}_2 = (1,0,1)$, $\mathbf{v}_3 = (1,-1,5)$
Answer: These vectors are linearly independent.

(b) $\mathbf{v}_1 = (2,6,3)$, $\mathbf{v}_2 = (1,-1,4)$, $\mathbf{v}_3 = (3,2,1)$, $\mathbf{v}_4 = (2,5,4)$
Answer: These vectors are linearly dependent. They are vectors in $\mathbb{R}^3$, which is a 3-dimensional vector space. Any set of more than 3 vectors in $\mathbb{R}^3$ is linearly dependent.

(c) $\mathbf{v}_1 = (1,-1,3,-1)$, $\mathbf{v}_2 = (1,-1,4,2)$, $\mathbf{v}_3 = (1,-1,5,7)$.
Answer: These vectors are linearly independent.

Show that in the space $\mathbb{R}^3$ the vectors $\mathbf{x} = (1,1,0)$, $\mathbf{y} = (0,1,2)$, and $\mathbf{z} = (3,1,-4)$ are linearly dependent by finding scalars α and β such that $\alpha\mathbf{x} + \beta\mathbf{y} + \mathbf{z} = \mathbf{0}$.

Answer: $\alpha =$ _____ , $\beta =$ _____ .

Solve(Any two)

Bases and Dimension of Subspaces in $\mathbb{R}^n$

Prove that the vectors $(1,1,0)$, $(1,2,3)$, and $(2,-1,5)$ form a basis for $\mathbb{R}^3$.

Find a basis for $\operatorname{Span}(S)$ where $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$.

Let

$$\mathbf{v}=\begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \mathbf{v}_1=\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2=\begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}.$$

Find the necessary and sufficient condition so that the vector $\mathbf{v}$ is a linear combination of the vectors $\mathbf{v}_1, \mathbf{v}_2$.

Find a basis of the null space of the matrix $B=\begin{bmatrix} 1 & 1 & 2 \\ -2 & -2 & -4 \end{bmatrix}$.

(a) Let $A=\begin{bmatrix} 1 & 3 & 0 & 0 \\ 1 & 3 & 1 & 2 \\ 1 & 3 & 1 & 2 \end{bmatrix}$. Find a basis for the range $\mathcal{R}(A)$ of A that consists of columns of A.

(b) Find the rank and nullity of the matrix A in part (a).

Find the values of λ that satisfy the equation

$$\begin{vmatrix} -3-\lambda & 10 \\ 2 & 5-\lambda \end{vmatrix} = 0.$$

Answer: $\lambda=-5, 7$

If $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 5$, evaluate the determinant of the matrix

$$\begin{bmatrix} 2a_1 & a_2 & a_3 \\ 6b_1 & 3b_2 & 3b_3 \\ 2c_1 & c_2 & c_3 \end{bmatrix}.$$

Use an inverse matrix to solve the linear system

$$\begin{aligned} x_1 + 2x_2 + 2x_3 &= 1 \\ x_1 - 2x_2 + 2x_3 &= -3 \\ 3x_1 - x_2 + 5x_3 &= 7 \end{aligned}$$

Answer: $x_1=21,\ x_2=1,\ x_3=-11$

Write the linear system

$$\begin{aligned} 7x_1 - 2x_2 &= b_1, \\ 3x_1 - 2x_2 &= b_2, \end{aligned}$$

in the form $A\mathbf{x}=\mathbf{b}$. Use $\mathbf{x}=A^{-1}\mathbf{b}$ to solve the system for each $\mathbf{b}$:

$$\mathbf{b}=\begin{bmatrix} 5 \\ 4 \end{bmatrix}, \ \mathbf{b}=\begin{bmatrix} 10 \\ 50 \end{bmatrix}, \ \mathbf{b}=\begin{bmatrix} 0 \\ -20 \end{bmatrix}.$$

Solve(Any three)

. Find the eigenvalues and eigenvectors of the given matrix.

(a) $\begin{bmatrix} -1 & 2 \\ -7 & 8 \end{bmatrix}$

Answer: $\lambda_1 = 1,\ \mathbf{v}_1 = (1,1),\ \lambda_2 = 6,\ \mathbf{v}_2 = (2,7)$

(b) $\begin{bmatrix} -1 & 2 \\ -5 & 1 \end{bmatrix}$

Answer: $\lambda_1 = 3i,\ \mathbf{v}_1 = (2, 1+3i),\ \lambda_2 = -3i,\ \mathbf{v}_2 = (2, 1-3i)$

(c) $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

Answer: $\lambda_1 = 1+i,\ \mathbf{v}_1 = (i,1),\ \lambda_2 = 1-i,\ \mathbf{v}_2 = (-i,1)$

(d) $\begin{bmatrix} 5 & -1 & 0 \\ 0 & -5 & 9 \\ 5 & -1 & 0 \end{bmatrix}$

Answer: $\lambda_1 = 4,\ \mathbf{v}_1 = (1,1,1),\ \lambda_2 = -4,\ \mathbf{v}_2 = (1,9,1),\ \lambda_3 = 0,\ \mathbf{v}_3 = (9,45,25)$

(e) $\begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 1 & 1 & -1 \end{bmatrix}$

Answer: $\lambda_1 = -1,\ \mathbf{v}_1 = (1,-1,1),\ \lambda_2 = i,\ \mathbf{v}_2 = (i,1,1),\ \lambda_3 = -i,\ \mathbf{v}_3 = (-i,1,1)$

(f) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & -7 \end{bmatrix}$

Answer: $\lambda_1 = 1,\ \mathbf{v}_1 = (1,0,0),\ \lambda_2 = 5,\ \mathbf{v}_2 = (1,2,0),\ \lambda_3 = -7,\ \mathbf{v}_3 = (1,2,-4)$

Choose a, b and c in the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a & b & c \end{bmatrix}$ so that the characteristic polynomial of A is $-\lambda^3 + 4\lambda^2 + 5\lambda + 6$.

Answer: $a =$ _____ ; $b =$ _____ ; and $c =$ _____ .

Suppose that it is known that the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ \sqrt{3} & a & 17 \\ 2 & 0 & b \end{bmatrix}$ has eigenvalues 2 and 3 and that the eigenvalue 2 has algebraic multiplicity 2. Then $a =$ _____ and $b =$ _____ .

Solve(Any two)

Determine whether the given matrix A is diagonalizable. If so, find the matrix P that diagonalizes A and the diagonal matrix D such that $D = P^{-1}AP$.

(a) $\begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$

Answer: This matrix has one eigenvalue with algebraic multiplicity 2, but only one linearly independent eigenvector. It is defective and therefore not diagonalizable.

(b) $\begin{bmatrix} -9 & 13 \\ -2 & 6 \end{bmatrix}$

Answer:

$$P = \begin{bmatrix} 1 & 13 \\ 1 & 2 \end{bmatrix}, \ D = \begin{bmatrix} 4 & 0 \\ 0 & -7 \end{bmatrix}$$

(c) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Answer:

$$P = \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}, \ D = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$$

(d) $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$

Answer:

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, \ D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

(e) $\begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

Answer: This matrix has two eigenvalues with algebraic multiplicities of 1 and 2, respectively. The latter has only one linearly independent eigenvector, hence the matrix is defective and not diagonalizable.

(f) $\begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$